

TRIGONOMETRIC EQUATIONS



SYNOPSIS

- Equations involving one or more than one trigonometric ratios of unknown angles are called trigonometric equations.

Ex: $3\cos^2\theta + 4\cos\theta = 0$ is trigonometric equation in unknown angle ' θ '.

- **Solution of a trigonometric equation :**

A value of the unknown angle satisfying the given equation is called the solution of the given equation. Every other coterminal angle of a solution is also solution of the given trigonometric equation.

- The trigonometric equation may have infinite number of solutions and can be classified as

(a) Principal solution (b) General solution

(a) **Principal solution:**

If θ has infinitely many solutions among them, the least value with sign is called principal value of θ . It is denoted by ' α '.

(i) There exists a unique value of ' θ ' in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

satisfying $\sin\theta = k, k \in R, |k| \leq 1$. This value of ' θ ' is called principal value of ' θ ' or principal solution of $\sin\theta = k$

(ii) There exists a unique value of ' θ ' in $[0, \pi]$

satisfying $\cos\theta = k, k \in R, |k| \leq 1$. This value of ' θ ' is called principal value of ' θ ' or principal solution of $\cos\theta = k$

(iii) There exists a unique value of ' θ ' in

$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying $\tan\theta = k, k \in R$. This value of ' θ ' is called principal value of ' θ ' or principal solution of $\tan\theta = k$.

(b) **General solution:** Since trigonometric functions are periodic, a solution generalised by means of periodicity of trigonometric functions. The solution consisting of all possible solutions of a trigonometric equation is called its general solution.

S n o	Equation	Interval in which principal solution α	General Solution
1	$\sin\theta = k,$ $(-1 \leq k \leq 1)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$\theta = n\pi + (-1)^n \alpha,$ $n \in Z$
2	$\cos\theta = k,$ $(-1 \leq k \leq 1)$	$[0, \pi]$	$\theta = 2n\pi \pm \alpha,$ $n \in Z$
3	$\tan\theta = k,$ $(k \in R)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$\theta = n\pi + \alpha,$ $n \in Z$
4	$\sec\theta = k,$ $k \in (-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$	$\theta = n\pi + (-1)^n \alpha,$ $n \in Z$
5	$\csc\theta = k,$ $k \in (-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$	$\theta = 2n\pi \pm \alpha,$ $n \in Z$
6	$\cot\theta = k,$ $k \in R$	$(0, \pi)$	$\theta = n\pi + \alpha,$ $n \in Z$

- The general solution of $\sin^2\theta = \sin^2\alpha$ (or) $\cos^2\theta = \cos^2\alpha$ (or) $\tan^2\theta = \tan^2\alpha$ is $\theta = n\pi \pm \alpha, n \in Z$

- **Common Solution of Two Equations:**
The general solution for the equations of the type $\sin\theta = \sin\alpha$, $\cos\theta = \cos\alpha$ is $\theta = 2n\pi + \alpha$ where, α is common angle lies between 0 and 2π

W.E-1: The solution of $\sin\theta = -\frac{1}{2}$, $\tan\theta = \frac{1}{\sqrt{3}}$ is

Sol: $\because \sin\theta < 0$ and $\tan\theta > 0 \Rightarrow \theta \in Q_3$,
 $\therefore$ principal common solution is $\pi + \pi/6 = 7\pi/6$
 $\therefore$ general solution $\theta = 2n\pi + \frac{7\pi}{6}, n \in Z$

- **Some important values to remember:**

Angle θ	Sin θ	Cos θ
15°	$\frac{\sqrt{3}-1}{2\sqrt{2}}$	$\frac{\sqrt{3}+1}{2\sqrt{2}}$
18°	$\frac{\sqrt{5}-1}{4}$	$\frac{\sqrt{10+2\sqrt{5}}}{4}$
$22\frac{1}{2}^\circ$	$\frac{\sqrt{2}-1}{2\sqrt{2}}$	$\frac{\sqrt{2}+1}{2\sqrt{2}}$
36°	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	$\frac{\sqrt{5}+1}{4}$

$$54^{\circ} \quad \frac{\sqrt{5}+1}{4} \quad \frac{\sqrt{10-2\sqrt{5}}}{4}$$

$$67\frac{1}{2}^{\circ} \quad \sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}} \quad \sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$$

$$72^{\circ} \quad \frac{\sqrt{10+2\sqrt{5}}}{4} \quad \frac{\sqrt{5}-1}{4}$$

$$75^{\circ} \quad \frac{\sqrt{3}+1}{2\sqrt{2}} \quad \frac{\sqrt{3}-1}{2\sqrt{2}}$$

- i) $\tan 15^{\circ} = 2 - \sqrt{3}$, $\tan 22\frac{1}{2}^{\circ} = \sqrt{2} - 1$,
 ii) $\cot 15^{\circ} = 2 + \sqrt{3}$, $\cot 22\frac{1}{2}^{\circ} = \sqrt{2} + 1$
- i) The equation $a \cos \theta + b \sin \theta = c$ has a solution if $|c| \leq \sqrt{a^2 + b^2}$.
 ii) The equation $a \cos \theta + b \sin \theta = c$ has no solution if $|c| > \sqrt{a^2 + b^2}$

➤ **Important Points to Remember:**

- (i) For equations of the type $\sin \theta = k$ or $\cos \theta = k$, one must check that $|k| \leq 1$.
 (ii) Avoid squaring the equation, if possible, because it may lead to extraneous solutions.
 (iii) Do not cancel the common variable factor from the two sides of the equations, because we may lose some solutions.
 (iv) The answer should not contain such values of θ which make any of the terms undefined or infinite.
 (v) Check that denominator is not zero at any stage while solving equations.
 (vi) (a) If $\tan \theta$ or $\sec \theta$ is involved then θ should not be an odd multiple of $\pi/2$.
 (b) If $\cot \theta$ or $\operatorname{cosec} \theta$ is involved then θ should not be a multiple of π or 0
 (vii) If two different trigonometric ratios, such as, $\tan \theta$ and $\sec \theta$ are involved then after solving we cannot apply the usual formulae for general solution, because periodicity of the functions are not same.
 (viii) If L.H.S. of the given trigonometric equation is always less than or equal to k and R.H.S. is always greater than or equal to k. If both the sides are equal to k for same value of θ , then solution exist and if they are equal for different value of θ , then solution does not exist.

➤ **Solution of Inequations:**

If $a < b$, then

- (i) $(x-a)(x-b) < 0 \Rightarrow a < x < b$
 (ii) $(x-a)(x-b) \leq 0 \Rightarrow a \leq x \leq b$
 (iii) $(x-a)(x-b) > 0 \Rightarrow x < a \text{ or } x > b$
 (iv) $(x-a)(x-b) \geq 0 \Rightarrow x \leq a \text{ or } x \geq b$



LEVEL - I

PRINCIPAL SOLUTION

1. If $4\cos^2 \theta = 3$ then $\theta = \dots\dots$
 1) $\frac{\pi}{6}, \frac{5\pi}{6}$ 2) $\frac{\pi}{4}, \frac{3\pi}{4}$ 3) $\frac{\pi}{3}, \frac{2\pi}{3}$ 4) $\pm \frac{\pi}{2}$
2. If $\tan \theta + \sec \theta = \sqrt{3}$, then the principal value of $\left(\theta + \frac{\pi}{6}\right)$ is
 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{2\pi}{3}$ 4) $\frac{\pi}{2}$
3. If $\tan^2 \theta = \sqrt{3} + (\sqrt{3} - 1)\tan \theta$ and θ lies in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ then $\theta =$
 1) $\left\{-\frac{\pi}{3}, \frac{\pi}{4}\right\}$ 2) $\left\{\frac{\pi}{3}, -\frac{\pi}{4}\right\}$
 3) $\left\{\frac{\pi}{6}, -\frac{\pi}{3}\right\}$ 4) $\left\{-\frac{\pi}{12}, \frac{\pi}{12}\right\}$
4. If $\sin\left(\frac{\pi \cot \theta}{4}\right) = \cos\left(\frac{\pi \tan \theta}{4}\right)$ and θ is in the first quadrant then $\theta = \dots\dots$
 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{6}$
5. If $3\tan^4 \alpha - 10\tan^2 \alpha + 3 = 0$ then principal values of ' α ' are
 1) $\pm 45^{\circ}, \pm 36^{\circ}$ 2) $\pm 30^{\circ}, \pm 60^{\circ}$
 3) $\pm 75^{\circ}, \pm 36^{\circ}$ 4) $\pm 60^{\circ}, \pm 15^{\circ}$
6. If $\sin(x + 28^{\circ}) = \cos(3x - 78^{\circ})$ then $x =$
 1) $37^{\circ}, 8^{\circ}$ 2) 39° 3) $35^{\circ}, 8^{\circ}$ 4) 47°
7. If $\sin 2\theta - \cos 3\theta = 0$ and θ is acute then $\theta =$
 1) 36° 2) 54° 3) 18° 4) 30°
8. The smallest value of θ satisfying the equation $\sqrt{3}(\cot \theta + \tan \theta) = 4$ is
 1) $\frac{2\pi}{3}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) $\frac{\pi}{12}$

TRIGONOMETRIC EQUATIONS

GENERAL SOLUTION

9. The most general value of θ satisfying the equations $\sin\theta = \frac{1}{\sqrt{2}}$, $\cos\theta = -\frac{1}{\sqrt{2}}$ is

1) $2n\pi + \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 2) $2n\pi + \frac{3\pi}{4}, \forall n \in \mathbb{Z}$
 3) $n\pi + \frac{\pi}{6}, \forall n \in \mathbb{Z}$ 4) $n\pi + \frac{\pi}{3}, \forall n \in \mathbb{Z}$

10. The general solution of $\frac{\tan 2x - \tan x}{1 + \tan x \tan 2x} = 1$ is

1) $n\pi + \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 2) $n\pi \pm \frac{\pi}{4}, \forall n \in \mathbb{Z}$
 3) ϕ 4) $n\pi + \frac{\pi}{6}, \forall n \in \mathbb{Z}$

11. If $11\sin^2 x + 7\cos^2 x = 8$ then $x =$ -----

1) $n\pi \pm \frac{\pi}{6}, \forall n \in \mathbb{Z}$ 2) $n\pi \pm \frac{\pi}{4}, \forall n \in \mathbb{Z}$
 3) $n\pi \pm \frac{\pi}{3}, \forall n \in \mathbb{Z}$ 4) $n\pi \pm \frac{\pi}{2}, \forall n \in \mathbb{Z}$

12. The general solution of $\sin 2x = 4\cos x$ is

1) $(2n+1)\frac{\pi}{2}, \forall n \in \mathbb{Z}$ 2) $n\pi, \forall n \in \mathbb{Z}$
 3) Empty Set 4) $2n\pi, \forall n \in \mathbb{Z}$

13. $\tan\left(\frac{\pi}{4} + \frac{A}{2}\right) + \tan\left(\frac{\pi}{4} - \frac{A}{2}\right) = \frac{4}{\sqrt{3}}$; then the general solution of $A =$

1) $2n\pi \pm \frac{\pi}{6}, \forall n \in \mathbb{Z}$ 2) $n\pi + \frac{\pi}{4}, \forall n \in \mathbb{Z}$
 3) $2n\pi \pm \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 4) $n\pi, \forall n \in \mathbb{Z}$

14. If $\tan x + 2\tan 2x + 4\tan 4x + 8\cot 8x = \sqrt{3}$ then the general solution of $x =$

1) $n\pi + \frac{\pi}{3}, \forall n \in \mathbb{Z}$ 2) $n\pi + \frac{\pi}{6}, \forall n \in \mathbb{Z}$
 3) $n\pi + \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 4) $n\pi, \forall n \in \mathbb{Z}$

15. The general value of θ satisfies the equation

$\tan\theta \tan(120^\circ + \theta) \tan(120^\circ - \theta) = \frac{1}{\sqrt{3}}$ is

1) $(6n+1)\frac{\pi}{18}, \forall n \in \mathbb{Z}$ 2) $(3n+1)\frac{\pi}{3}, \forall n \in \mathbb{Z}$
 3) $(6n+1)\frac{\pi}{6}, \forall n \in \mathbb{Z}$ 4) $(3n+1)\frac{\pi}{6}, \forall n \in \mathbb{Z}$

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16. If $\tan m\theta = \cot n\theta$ then the G.S of $\theta =$

1) $\frac{(k+1)\pi}{2(m+n)}, \forall k \in \mathbb{Z}$ 2) $\frac{(2k+1)\pi}{2(m+n)}, \forall k \in \mathbb{Z}$

3) $\frac{(2k+1)\pi}{m+n}, \forall k \in \mathbb{Z}$ 4) $\frac{(2k+1)\pi}{m-n}, \forall k \in \mathbb{Z}$

17. If $\cos x + \cos y = 1$ and $\cos x \cos y = 1/4$ then the G.S are

1) $x = 2n\pi \pm \frac{\pi}{4}, n \in \mathbb{Z}, y = 2m\pi \pm \frac{\pi}{4}, m \in \mathbb{Z}$

2) $x = 2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}, y = 2m\pi \pm \frac{\pi}{3}, m \in \mathbb{Z}$

3) $x = 2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}, y = 2m\pi \pm \frac{\pi}{6}, m \in \mathbb{Z}$

4) $x = 2n\pi \pm \frac{\pi}{5}, n \in \mathbb{Z}, y = 2m\pi \pm \frac{\pi}{5}, m \in \mathbb{Z}$

18. If $\sin x \cdot \sin(60^\circ + x) \cdot \sin(60^\circ - x) = \frac{1}{8}$, then $x =$

1) $n\pi + (-1)^n \frac{\pi}{6}$ 2) $\frac{n\pi}{3} + (-1)^n \frac{\pi}{18}$

3) $n\pi + (-1)^n \frac{\pi}{3}$ 4) $\frac{n\pi}{3} + (-1)^n \frac{\pi}{9}$

SOLUTIONS IN THE GIVEN INTERVAL

19. The value of θ satisfying

$\sin 7\theta = \sin 4\theta - \sin \theta$ in $0 < \theta < \frac{\pi}{2}$ are

1) $\frac{\pi}{9}, \frac{\pi}{4}$ 2) $\frac{\pi}{3}, \frac{\pi}{9}$ 3) $\frac{\pi}{6}, \frac{\pi}{4}$ 4) 0, 1

20. If $1 + \sin^2 \theta = 3\sin \theta \cos \theta$ then the solution set in $\left(0, \frac{\pi}{2}\right)$ is

1) $\left\{\frac{\pi}{4}, \cos^{-1} \frac{1}{3}\right\}$ 2) $\left\{\frac{\pi}{4}, \tan^{-1} \frac{1}{2}\right\}$

3) $\left\{\frac{\pi}{3}, \tan^{-1} \frac{1}{3}\right\}$ 4) $\left\{\frac{\pi}{6}, \sin^{-1} \frac{1}{3}\right\}$

21. The equation $\sqrt{3}\sin x + \cos x = 4$ has

1) Only one solution 2) Two solutions
 3) Infinitely many solutions 4) No solution

22. If $y + \cos \theta = \sin \theta$ has a real solution then

1) $-\sqrt{2} \leq y \leq \sqrt{2}$ 2) $y > \sqrt{2}$

3) $y \leq -\sqrt{2}$ 4) $y \leq 1$

LEVEL-I-KEY

01) 1 02) 1 03) 2 04) 3 05) 2 06) 3
07) 3 08) 3 09) 2 10) 3 11) 1 12) 1
13) 1 14) 2 15) 1 16) 2 17) 2 18) 2
19) 1 20) 2 21) 4 22) 1

LEVEL-I-HINTS

- $4 \cos^2 \theta = 3$
 $\Rightarrow \cos^2 \theta = \left(\frac{\sqrt{3}}{2}\right)^2 \Rightarrow \theta = n\pi \pm \frac{\pi}{6}$
- $\tan \theta + \sec \theta = \sqrt{3} \Rightarrow \sin \theta + 1 = \sqrt{3} \cos \theta$
 $\Rightarrow \sqrt{3} \cos \theta - \sin \theta = 1 \Rightarrow \frac{\sqrt{3}}{2} \cos \theta - \frac{1}{2} \sin \theta = \frac{1}{2}$
 $\Rightarrow \cos \theta \cos \frac{\pi}{6} - \sin \theta \sin \frac{\pi}{6} = \frac{1}{2} \Rightarrow \cos \left(\theta + \frac{\pi}{6}\right) = \frac{1}{2}$
 $\therefore$ P.S. of $\theta + \frac{\pi}{6} = \frac{\pi}{3}$
- $\tan^2 \theta = \sqrt{3} + (\sqrt{3} - 1) \tan \theta \quad \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
on verification $\left\{\frac{\pi}{3}, -\frac{\pi}{4}\right\}$ satisfies
- $\sin \left(\frac{\pi \cot \theta}{4}\right) = \cos \left(\frac{\pi \tan \theta}{4}\right)$
on verification $\theta = \frac{\pi}{4}$ satisfies
- $3 \tan^4 \alpha - 10 \tan^2 \alpha + 3 = 0$
 $\Rightarrow 3 \tan^4 \alpha - 9 \tan^2 \alpha - \tan^2 \alpha + 3 = 0$
 $\Rightarrow (3 \tan^2 \alpha - 1)(\tan^2 \alpha - 3) = 0$
 $\Rightarrow \tan^2 \alpha = \frac{1}{3} = \left(\frac{1}{\sqrt{3}}\right)^2; \tan^2 \alpha = 3 = (\sqrt{3})^2$
 $\therefore \alpha = \pm 30^\circ, \pm 60^\circ$
- $3x - 78^\circ = 90^\circ - (x + 28^\circ) \text{ or } 90 + x + 28^\circ$
 $4x = 90^\circ - 28^\circ + 78^\circ = 140^\circ \Rightarrow x = \frac{140^\circ}{4} = 35^\circ$
 $x = 8^\circ$
- $\sin 2\theta = \cos 3\theta$
 $3\theta = \frac{\pi}{2} - 2\theta \Rightarrow 5\theta = 90^\circ \Rightarrow \theta = 18^\circ$
- $2 \operatorname{cosec} 2\theta = \frac{4}{\sqrt{3}}$
 $\sin 2\theta = \frac{\sqrt{3}}{2} \Rightarrow 2\theta = \frac{\pi}{3}; \theta = \frac{\pi}{6}$

- $\sin \theta = \frac{1}{\sqrt{2}}; \cos \theta = -\frac{1}{\sqrt{2}}$
 $\therefore \theta \in \text{II quadrant}$
 $\therefore$ P.S. of $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$
G.S. of $\theta = 2n\pi + \frac{3\pi}{4}, \forall n \in \mathbb{Z}$
- $\tan(3x - 2x) = 1 \Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4}$
But for $x = \frac{\pi}{4}, 2x = \frac{\pi}{2} \Rightarrow \tan 2x$ is undefined
 $\therefore \exists$ no solutions
- $11 \sin^2 x + 7 \cos^2 x = 8 \Rightarrow 4 \sin^2 x + 7 = 8$
 $\Rightarrow 4 \sin^2 x = 1 \Rightarrow \sin^2 x = \frac{1}{4} = \left(\frac{1}{2}\right)^2$
 $\therefore x = n\pi \pm \frac{\pi}{6}$
- $\sin 2x = 4 \cos x \Rightarrow 2 \sin x \cos x = 4 \cos x$
 $\Rightarrow \cos x = 0$ or $\sin x = 2$ (which is not possible), $\therefore x = (2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}$
- $\tan\left(\frac{\pi}{4} + \frac{A}{2}\right) + \tan\left(\frac{\pi}{4} - \frac{A}{2}\right) = \frac{4}{\sqrt{3}}$
 $\Rightarrow \tan\left(\frac{\pi}{4} + \frac{A}{2}\right) + \cot\left(\frac{\pi}{4} + \frac{A}{2}\right) = \frac{4}{\sqrt{3}}$
 $\Rightarrow 2 \operatorname{cosec}\left(2\left(\frac{\pi}{4} + \frac{A}{2}\right)\right) = \frac{4}{\sqrt{3}} (\because \tan \theta + \cot \theta = 2 \operatorname{cosec} 2\theta)$
 $\Rightarrow 2 \operatorname{cosec}\left(\frac{\pi}{2} + A\right) = \frac{4}{\sqrt{3}} \Rightarrow \sec A = \frac{2}{\sqrt{3}} \Rightarrow \cos A = \frac{\sqrt{3}}{2}$
 $\Rightarrow A = 2n\pi \pm \frac{\pi}{6}, \forall n \in \mathbb{Z}$
- $\tan x + 2 \tan 2x + 4 \tan 4x + 8 \cot 8x = \sqrt{3}$
 $\cot x - 8 \cot 8x + 8 \cot 8x = \sqrt{3} \Rightarrow \cot x = \sqrt{3}$
 $\Rightarrow x = n\pi + \frac{\pi}{6}, n \in \mathbb{Z}$

TRIGONOMETRIC EQUATIONS

15. $\tan \theta \tan (120^\circ + \theta) \tan (120^\circ - \theta) = \frac{1}{\sqrt{3}}$
 $\Rightarrow \tan 3\theta = \frac{1}{\sqrt{3}} \Rightarrow 3\theta = n\pi + \frac{\pi}{6} \Rightarrow \theta = \frac{n\pi}{3} + \frac{\pi}{18}$
 $= \left(n + \frac{1}{6}\right) \frac{\pi}{3} = (6n+1) \frac{\pi}{18}$
16. $\tan m\theta = \cot n\theta = \tan \left(\frac{\pi}{2} - n\theta\right)$
 $m\theta = K\pi + \frac{\pi}{2} - n\theta \Rightarrow \theta = \frac{K\pi + \frac{\pi}{2}}{m+n} = \frac{(2K+1)\pi}{2(m+n)}$
17. $\cos x + \cos y = 1; \cos x \cos y = \frac{1}{4}$
 on verification $x = y = \frac{\pi}{3}$ satisfies
18. $\frac{1}{4} \sin 3x = \frac{1}{8} \Rightarrow 3x = n\pi + (-1)^n \frac{\pi}{6}$
19. $\sin 7\theta + \sin \theta = \sin 4\theta$
 $2 \sin 4\theta \cdot \cos 3\theta = \sin 4\theta$
 $\sin 4\theta = 0, \quad \cos 3\theta = \frac{1}{2}$
 $\theta = \frac{\pi}{4} \quad 3\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{9}$
20. Divide both sides with $\cos^2 \theta$
 $\sec^2 \theta + \tan^2 \theta = 3 \tan \theta$
 $\Rightarrow 2 \tan^2 \theta - 3 \tan \theta + 1 = 0$
 $\Rightarrow (2 \tan \theta - 1)(\tan \theta - 1) = 0$
 $\Rightarrow \tan \theta = 1 \quad \text{or} \quad \tan \theta = \frac{1}{2}$
 $\Rightarrow \theta = \frac{\pi}{4} \quad \text{or} \quad \theta = \tan^{-1} \left(\frac{1}{2}\right)$
21. $\sqrt{3} \sin x + \cos x = 4$ (no solution)
 $[a \cos x + b \sin x = c \quad \text{has a solution}$
 if $-\sqrt{a^2 + b^2} \leq c \leq \sqrt{a^2 + b^2}]$
22. $y = \sin \theta - \cos \theta$ It has solutions if
 $|c| \leq \sqrt{a^2 + b^2} \Rightarrow -\sqrt{2} \leq y \leq \sqrt{2}$

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LEVEL - I

1. If $3 \tan \theta = \cot \theta$ then $\theta = \dots$
 1) $\pm 30^\circ$ 2) $\pm 60^\circ$ 3) $\pm 45^\circ$ 4) $\pm 15^\circ$
2. The principal value of $\left(\theta + \frac{\pi}{4}\right)$ where
 $\sin \theta + \cos \theta = 1$ is
 1) 0° 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{2}$
3. If $\cos 2\theta = 2 \sin^2 \theta$ then $\theta =$
 1) $\pm 30^\circ$ 2) $\pm 60^\circ$ 3) $\pm 45^\circ$ 4) $\pm 90^\circ$
4. If $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$ then the value(s)
 of $\cos\left(\theta - \frac{\pi}{4}\right)$ is (are)
 1) $\frac{1}{2}$ 2) $\frac{1}{\sqrt{2}}$ 3) $\pm \frac{1}{2\sqrt{2}}$ 4) $\frac{1}{3\sqrt{2}}$
5. If A and B are acute angles such that
 $\sin A = \sin^2 B$ and $2 \cos^2 A = 3 \cos^2 B$ then A =
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$
6. If $\sqrt{3} \cos \theta - \sin \theta = 1$ then $\theta =$
 1) π 2) $\frac{\pi}{2}$ 3) 1 4) $\frac{\pi}{6}$
7. If $\cot \theta - \tan \theta = 2$ then principal value of θ
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{8}$ 4) $\frac{3\pi}{4}$
8. If $\sin A = \sin B, \cos A = \cos B$ then A =
 1) $n\pi + B, \forall n \in \mathbb{Z}$ 2) $2n\pi + B, \forall n \in \mathbb{Z}$
 3) $2n\pi - B, \forall n \in \mathbb{Z}$ 4) $2n\pi \pm B, \forall n \in \mathbb{Z}$
9. The general solution of $\frac{\tan 5x - \tan 4x}{1 + \tan 5x \tan 4x} = 1$ is
 1) $n\pi + \frac{\pi}{4}; \forall n \in \mathbb{Z}$ 2) $n\pi \pm \frac{\pi}{4}, \forall n \in \mathbb{Z}$
 3) ϕ 4) $n\pi + \frac{\pi}{6}, \forall n \in \mathbb{Z}$
10. Solution of $\cot^2 \theta + \left(\sqrt{3} + \frac{1}{\sqrt{3}}\right) \cot \theta + 1 = 0$ is
 1) $n\pi - \frac{\pi}{6}, n\pi - \frac{\pi}{3}, \forall n \in \mathbb{Z}$
 2) $n\pi + \frac{\pi}{6}, n\pi + \frac{\pi}{3}, \forall n \in \mathbb{Z}$
 3) $n\pi + \frac{\pi}{12}, \forall n \in \mathbb{Z}$ 4) $n\pi + \frac{\pi}{4}, \forall n \in \mathbb{Z}$

11. $3\sin x + 4\cos x - 6 = 0$ then the general solution of $x =$

- 1) $n\pi + (-1)^n \frac{\pi}{6}, \forall n \in \mathbb{Z}$
 2) $n\pi + (-1)^n \frac{\pi}{4}, \forall n \in \mathbb{Z}$
 3) $n\pi + (-1)^n \frac{\pi}{3}, \forall n \in \mathbb{Z}$ 4) Empty Set

12. $\tan\left(\frac{\pi}{4} + \frac{x}{2}\right) - \tan\left(\frac{\pi}{4} - \frac{x}{2}\right) = 2$ then the general solution of $x =$

- 1) $n\pi \pm \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 2) $n\pi + \frac{\pi}{4}, \forall n \in \mathbb{Z}$
 3) $n\pi - \frac{\pi}{4}, \forall n \in \mathbb{Z}$ 4) $n\pi, \forall n \in \mathbb{Z}$

13. If $\tan A + \tan 2A + \sqrt{3} \tan A \tan 2A = \sqrt{3}$ then the general solution of $\frac{A}{2} =$

- 1) $\frac{n\pi}{3} + \frac{\pi}{9}, \forall n \in \mathbb{Z}$ 2) $n\pi + \frac{\pi}{9}, \forall n \in \mathbb{Z}$
 3) $\frac{n\pi}{6} + \frac{\pi}{18}, \forall n \in \mathbb{Z}$ 4) $\frac{n\pi}{3} + \frac{\pi}{4}, \forall n \in \mathbb{Z}$

14. Solution of $\tan x + \tan(120^\circ + x) + \tan(x - 120^\circ) = 0$ is

- 1) $\frac{n\pi}{3}, \forall n \in \mathbb{Z}$ 2) $\frac{n\pi}{6}, \forall n \in \mathbb{Z}$
 3) $\frac{n\pi}{4}, \forall n \in \mathbb{Z}$ 4) $\frac{n\pi}{2}, \forall n \in \mathbb{Z}$

15. If $\tan p \theta = \cot q \theta$ then solutions of θ are in -
 1) A.P 2) G.P 3) H.P 4) A.G.P

16. If $\sin x + \cos x = \sqrt{2} \cos \alpha \Rightarrow x =$

- 1) $2n\pi - \frac{\pi}{4} \pm \alpha$ 2) $n\pi - \frac{\pi}{4} + \alpha$
 3) $2n\pi + \frac{\pi}{4} \pm \alpha$ 4) $n\pi - \frac{\pi}{4} \pm \alpha$

17. The solution of $(\sin x - 2)^2 + (\cos x - 3/2)^2 = 0$ is
 1) $x = \pi/2$ 2) $x = \pi$ 3) $x = \pi/4$ 4) No solution

18. The values of θ satisfying

$\sin 5\theta = \sin 3\theta - \sin \theta$ and $0 < \theta < \frac{\pi}{2}$ are

- 1) $\frac{\pi}{6}, \frac{\pi}{3}$ 2) $\frac{\pi}{6}, \frac{\pi}{4}$ 3) $\frac{\pi}{4}, \frac{\pi}{3}$ 4) $\frac{\pi}{4}, \frac{\pi}{2}$

19. If $5\cos 2\theta + 2\cos^2(\theta/2) + 1 = 0, 0 < \theta < \pi \Rightarrow \theta =$

- 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{3}, \cos^{-1}\left(\frac{3}{5}\right)$
 3) $\cos^{-1}\left(\frac{3}{5}\right)$ 4) $\frac{\pi}{3}, \pi - \cos^{-1}\left(\frac{3}{5}\right)$

20. The number of solutions of the equation

$\sin 3\alpha = 4\sin \alpha \cdot \sin(x + \alpha) \cdot \sin(x - \alpha)$

where $0 < \alpha < \pi$ for x in the interval $[0, \pi]$ is

- 1) 1 2) 2 3) 4 4) 5

21. General solution of $2^{\sin x} + 2^{\cos x} = 2^{1-\frac{1}{\sqrt{2}}}$ is

- 1) $n\pi - \frac{\pi}{4}$ 2) $2n\pi + \frac{5\pi}{4}$
 3) $n\pi + (-1)^n \frac{\pi}{4}$ 4) $2n\pi \pm \frac{\pi}{4}$

LEVEL - I - KEY

- 01) 1 02) 3 03) 1 04) 3 05) 2 06) 4
 07) 3 08) 2 09) 1 10) 1 11) 4 12) 2
 13) 3 14) 1 15) 1 16) 3 17) 4 18) 1
 19) 4 20) 2 21) 2

LEVEL - I - HINTS

1. $3 \tan \theta = \cot \theta \Rightarrow \tan^2 \theta = \frac{1}{3} = \left(\frac{1}{\sqrt{3}}\right)^2$
 $\Rightarrow \theta = n\pi \pm \frac{\pi}{6}$ according to options $\pm 30^\circ$
2. $\sin \theta + \cos \theta = 1$
 $\Rightarrow \frac{1}{\sqrt{2}} \sin \theta + \frac{1}{\sqrt{2}} \cos \theta = \frac{1}{\sqrt{2}}$
 $\Rightarrow \sin\left(\theta + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}, \therefore$ P.S of $\theta + \frac{\pi}{4} = \frac{\pi}{4}$
3. $1 - 2 \sin^2 \theta = 2 \sin^2 \theta, 4 \sin^2 \theta = 1$
 $\sin \theta = \pm \frac{1}{2}, \theta = \pm 30^\circ$
4. $\tan(\pi \cos \theta) = \cot(\pi \sin \theta)$
 $\pi \cos \theta + \pi \sin \theta = \frac{\pi}{2} \Rightarrow \cos \theta + \sin \theta = \frac{1}{2}$
 $\Rightarrow \frac{1}{\sqrt{2}} \cos \theta + \frac{1}{\sqrt{2}} \sin \theta = \frac{1}{2\sqrt{2}} \Rightarrow \cos\left(\theta - \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}}$
5. $\sin A = \sin^2 B, 2 \cos^2 A = 3 \cos^2 B$
 $\Rightarrow 2 \cos^2 A = 3(1 - \sin A)$
 $\Rightarrow 2 - 2 \sin^2 A - 3 + 3 \sin A = 0$
 $\Rightarrow 2 \sin^2 A - 3 \sin A + 1 = 0$
 $\Rightarrow 2 \sin^2 A - 2 \sin A - \sin A + 1 = 0$
 $\Rightarrow (2 \sin A - 1)(\sin A - 1) = 0$
 $\Rightarrow \sin A = \frac{1}{2}, \therefore A = \frac{\pi}{6} (\because A \text{ is acute})$

TRIGONOMETRIC EQUATIONS

6. $\theta = \frac{\pi}{6}$ satisfies
7. $\cot \theta - \tan \theta = 2$ on verification $\theta = \frac{\pi}{8}$ satisfies
 $\left(\cot \frac{\pi}{8} = \sqrt{2} + 1; \tan \frac{\pi}{8} = \sqrt{2} - 1 \right)$
8. $\sin A - \sin B = 0; \cos A - \cos B = 0$
 $2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right) = 0; -2 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right) = 0$
 $\therefore \sin \left(\frac{A-B}{2} \right) = 0 \Rightarrow \frac{A-B}{2} = n\pi \Rightarrow A = 2n\pi + B$
9. $\tan(5x - 4x) = 1$
 $\Rightarrow \tan x = 1 \Rightarrow x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$
10. $\cot^2 \theta + \left(\sqrt{3} + \frac{1}{\sqrt{3}} \right) \cot \theta + 1 = 0$
 $\Rightarrow \cot^2 \theta + \sqrt{3} \cot \theta + \frac{1}{\sqrt{3}} \cot \theta + 1 = 0$
 $\Rightarrow \cot \theta \left(\cot \theta + \sqrt{3} \right) + \frac{1}{\sqrt{3}} \left(\cot \theta + \sqrt{3} \right) = 0$
 $\Rightarrow \left(\cot \theta + \sqrt{3} \right) \left(\cot \theta + \frac{1}{\sqrt{3}} \right) = 0$
 $\Rightarrow \cot \theta = -\sqrt{3}; \cot \theta = -\frac{1}{\sqrt{3}}$
 $\Rightarrow \tan \theta = -\frac{1}{\sqrt{3}}; \tan \theta = -\sqrt{3}$
 $\theta = n\pi - \frac{\pi}{6}; n\pi - \frac{\pi}{3}$
11. $3 \sin x + 4 \cos x = 6$, Here $|c| > \sqrt{a^2 + b^2}$
 $\therefore$ There exists no solution
12. $\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) - \tan \left(\frac{\pi}{4} - \frac{x}{2} \right) = 2 \tan x$
 $\therefore 2 \tan x = 2 \Rightarrow \tan x = 1$
 $\Rightarrow x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$
13. $\tan A + \tan 2A = \sqrt{3} (1 - \tan A \tan 2A)$
 $\Rightarrow \frac{\tan 2A + \tan A}{1 - \tan 2A \tan A} = \sqrt{3} \Rightarrow \tan 3A = \sqrt{3}$
 $\Rightarrow 3A = n\pi + \frac{\pi}{3} \Rightarrow A = \frac{n\pi}{3} + \frac{\pi}{9} \Rightarrow \frac{A}{2} = \frac{n\pi}{6} + \frac{\pi}{18}$

JEE-MAINS ADVANCED

14. $\tan x + \tan(120^\circ + x) + \tan(x - 120^\circ) = 0$
 $\Rightarrow 3 \tan 3x = 0 \Rightarrow \tan 3x = 0$
 $\Rightarrow 3x = n\pi \Rightarrow x = \frac{n\pi}{3}$
15. $\tan p\theta = \tan \left(\frac{\pi}{2} - q\theta \right)$, The solutions are in A.P
16. $\cos \left(x - \frac{\pi}{4} \right) = \cos \alpha$
 $\Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \alpha \Rightarrow x = 2n\pi + \frac{\pi}{4} \pm \alpha$
17. $\sin x - 2 = 0, \cos x - 3/2 = 0$
 but $\sin x \neq 2, \cos x \neq \frac{3}{2}$
18. $\sin 5\theta + \sin \theta = \sin 3\theta$
 $\Rightarrow 2 \sin 3\theta \cdot \cos 2\theta = \sin 3\theta$
 $\Rightarrow \sin 3\theta = 0$ or $\cos 2\theta = \frac{1}{2}, \frac{\pi}{6}, \frac{\pi}{3}$ satisfies
19. $5(2 \cos^2 \theta - 1) + 2 \left(\frac{1 + \cos \theta}{2} \right) + 1 = 0$
 $10 \cos^2 \theta - 5 \cos \theta + 6 \cos \theta - 3 = 0$
 $(5 \cos \theta + 3)(2 \cos \theta - 1) = 0$
 $\cos \theta = -\frac{3}{5}, \cos \theta = \frac{1}{2}, \theta = \frac{\pi}{3}, \pi - \cos^{-1} \left(\frac{3}{5} \right)$
20. $3 \sin \alpha - 4 \sin^3 \alpha = 4 \sin \alpha (\sin^2 x - \sin^2 \alpha)$
 $\Rightarrow 3 \sin \alpha - 4 \sin^3 \alpha = 4 \sin \alpha \sin^2 x - 4 \sin^3 \alpha$
 $\Rightarrow 4 \sin \alpha \sin^2 x = 3 \sin \alpha \Rightarrow \sin^2 x = \frac{3}{4} = \left(\frac{\sqrt{3}}{2} \right)^2$
 $\Rightarrow x = n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z} \quad x = \left\{ \frac{\pi}{6}, \frac{5\pi}{6} \right\}$
21. The given equality holds for minimum of $\sin x + \cos x$ i.e., $-\sqrt{2}$
 $\therefore x = \frac{5\pi}{4}$ G.S is $2n\pi + \frac{5\pi}{4}$



LEVEL - II

1. If $\cos 2\theta \cdot \cos 3\theta \cdot \cos \theta = \frac{1}{4}$ for
 $0 < \theta < \pi$, then $\theta =$
- 1) $\frac{\pi}{7}, \frac{5\pi}{7}, \pi$ 2) $\frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{6}, \frac{5\pi}{6}$
- 3) $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{\pi}{3}, \frac{2\pi}{3}$ 4) $\frac{\pi}{9}, \frac{4\pi}{9}, \frac{5\pi}{9}$

2. If α and β are two different solutions lying between $\frac{-\pi}{2}$ and $\frac{\pi}{2}$ of the equation

$2\tan\theta + \sec\theta = 2$ then $\tan\alpha + \tan\beta$ is

- 1) 0 2) 1 3) $\frac{4}{3}$ 4) $\frac{8}{3}$

3. If $\gamma \sin\theta = 3$, $\gamma = 4(1 + \sin\theta)$, $0 \leq \theta \leq 2\pi$ then $\theta =$ ----

- 1) $\frac{\pi}{6}, \frac{5\pi}{6}$ 2) $\frac{\pi}{3}, \frac{2\pi}{3}$ 3) $\frac{\pi}{4}, \frac{5\pi}{4}$ 4) $\frac{\pi}{2}, \pi$

4. If $32\tan^8\theta = 2\cos^2\alpha - 3\cos\alpha$ and $3\cos 2\theta = 1$ then the general value of ' α ' is

- 1) $n\pi \pm \frac{\pi}{3}$ 2) $2n\pi \pm \frac{2\pi}{3}$ 3) $n\pi \pm \pi$ 4) $2n\pi \pm \frac{\pi}{2}$

5. If $\cos 20^\circ = k$ and $\cos x = 2k^2 - 1$, then the possible values of x between 0° and 360° are

- 1) 140° 2) 40° and 140°
3) 40° and 320° 4) 50° and 130°

6. The equation $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$, where x is a variable with real roots. then the interval of p may be any one of the following.

- 1) $(0, 2\pi)$ 2) $(-\pi, 0)$ 3) $(-\frac{\pi}{2}, \frac{\pi}{2})$ 4) $(0, \pi]$

7. In a right angled triangle, the hypotenuse is $2\sqrt{2}$ times the length of the perpendicular drawn from the opposite vertex to the hypotenuse. Then the other two angles are

- 1) $\frac{\pi}{3}, \frac{\pi}{6}$ 2) $\frac{\pi}{4}, \frac{\pi}{4}$ 3) $\frac{\pi}{8}, \frac{3\pi}{8}$ 4) $\frac{\pi}{12}, \frac{5\pi}{12}$

8. Number of solutions of the equation $\tan x + \sec x = 2 \cos x$ in the interval $[0, 2\pi]$ is

- 1) 0 2) 1 3) 2 4) 3

9. The number of integral values of k for which the equation $7 \cos x + 5 \sin x = 2k + 1$ has a solution is

- 1) 4 2) 8 3) 10 4) 12

10. The number of distinct real roots of

$$\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0 \text{ in the interval}$$

$$-\frac{\pi}{4} \leq x \leq \frac{\pi}{4} \text{ is}$$

- 1) 0 2) 2 3) 1 4) 3

11. If the solutions for θ of $\cos p\theta + \cos q\theta = 0$, $p > q > 0$ are in A.P then numerically smallest common difference of A.P is

- 1) $\frac{\pi}{p+q}$ 2) $\frac{2\pi}{p+q}$ 3) $\frac{\pi}{2(p+q)}$ 4) $\frac{1}{p+q}$

12. The solution set of $\tan(4k+2)x - \tan(4k+1)x - \tan(4k+2)x \tan(4k+1)x = 1$, $k \in \mathbb{Z}$ is

- 1) ϕ 2) $\frac{\pi}{4}$
3) $\left\{n\pi + \frac{\pi}{4} : n \in \mathbb{Z}\right\}$ 4) $\left\{2n\pi + \frac{\pi}{4} : n \in \mathbb{Z}\right\}$

13. The general solution of the equation

$$\sin^2\theta \sec\theta + \sqrt{3}\tan\theta = 0 \text{ is}$$

- 1) $\theta = n\pi + (-1)^n \frac{\pi}{3}, \forall n \in \mathbb{Z}$
2) $\theta = n\pi, \forall n \in \mathbb{Z}$
3) $\theta = n\pi + (-1)^n \frac{\pi}{4}, \forall n \in \mathbb{Z}$
4) $\theta = \frac{n\pi}{6}, \forall n \in \mathbb{Z}$

14. $\cos^3\alpha + \cos^3(120^\circ + \alpha) + \cos^3(120^\circ - \alpha) = \frac{3\sqrt{3}}{4}$ then the general solution of α is

- 1) ϕ 2) $2n\pi \pm \frac{\pi}{3}, \forall n \in \mathbb{Z}$
3) $(2n+1)\frac{\pi}{2}, \forall n \in \mathbb{Z}$ 4) $n\pi, \forall n \in \mathbb{Z}$

15. The general solution of $\sin^{100}x - \cos^{100}x = 1$ is

- 1) $2n\pi + \frac{\pi}{3}, n \in \mathbb{I}$ 2) $n\pi + \frac{\pi}{4}, n \in \mathbb{I}$
3) $n\pi + \frac{\pi}{2}, n \in \mathbb{I}$ 4) $2n\pi - \frac{\pi}{3}, n \in \mathbb{I}$

16. The equation $4\sin^2x + 4\sin x + a^2 - 3 = 0$ has a solution if

- 1) $-2 \leq a \leq 2$ 2) $-1 \leq a \leq 1$
3) $-3 \leq a \leq 3$ 4) $-4 \leq a \leq 4$

17. If the complex numbers $\sin x + i \cos 2x$ and $\cos x - i \sin 2x$ are conjugate to each other, then the set of values of $x =$

- 1) $n\pi$ 2) $(2n+1)\frac{\pi}{2}$ 3) $\{0\}$ 4) ϕ

TRIGONOMETRIC EQUATIONS

18. Given that $\tan A, \tan B$ are the roots of the equation $x^2 - bx + c = 0$, the value of $\sin^2(A+B)$ is

1) $\frac{b}{(b+c)^2}$ 2) $\frac{b^2}{b^2+c^2}$
 3) $\frac{b^2}{c^2+(1-b^2)}$ 4) $\frac{b^2}{b^2+(1-c)^2}$

19. The smallest positive x satisfying

$$\log_{\cos x} \sin x + \log_{\sin x} \cos x = 2 \text{ is}$$

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{6}$

20. The number of solutions of the equation

$$2^{\cos x} = |\sin x| \text{ in } [-2\pi, 2\pi]$$

1) 1 2) 4 3) 3 4) 8

21. If the equation $k \cos x - 3 \sin x = k+1$ has a solution for 'x' then

1) $K \leq 4$ 2) $K \geq 4$
 3) $1 \leq k \leq 6$ 4) $0 \leq k \leq 8$

LEVEL - II - KEY

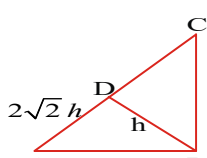
01) 3 02) 4 03) 1 04) 2 05) 3 06) 4
 07) 3 08) 3 09) 2 10) 3 11) 2 12) 1
 13) 2 14) 1 15) 3 16) 1 17) 4 18) 4
 19) 3 20) 4 21) 1

LEVEL - II - HINTS

1. $4 \cos 2\theta \cos 3\theta \cos \theta = 1$ $0 < \theta < \pi$
 $2 \cos 2\theta (\cos 4\theta + \cos 2\theta) - 1 = 0$
 $\Rightarrow 2 \cos 4\theta \cos 2\theta + 2 \cos^2 2\theta - 1 = 0$
 $\Rightarrow 2 \cos 4\theta \cos 2\theta + \cos 4\theta = 0$
 $\Rightarrow \cos 4\theta = 0$ or $\cos 2\theta = -\frac{1}{2}$
 $4\theta = (2n+1)\frac{\pi}{2}, 2\theta = 2n\pi \pm \frac{2\pi}{3}$
 $\Rightarrow \theta = (2n+1)\frac{\pi}{8}, \theta = n\pi \pm \frac{\pi}{3}$
 $\therefore \theta = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}; \theta = \frac{\pi}{3}, \frac{2\pi}{3}$
 2. $\sec \theta = 2(1 - \tan \theta)$
 $1 + \tan^2 \theta = 4(1 - 2 \tan \theta + \tan^2 \theta)$
 $\Rightarrow 3 \tan^2 \theta - 8 \tan \theta + 3 = 0 \Rightarrow \tan \alpha + \tan \beta = \frac{8}{3}$

JEE-MAINS ADVANCED

3. $\gamma \sin \theta = 3, \gamma = 4(1 + \sin \theta), 0 \leq \theta \leq 2\pi$
 on verification with the options $\frac{\pi}{6}, \frac{5\pi}{6}$ satisfies
 4. $3 \cos 2\theta = 1 \Rightarrow \frac{3(1 - \tan^2 \theta)}{1 + \tan^2 \theta} = 1 \Rightarrow \tan^2 \theta = \frac{1}{2}$
 $32\left(\frac{1}{2}\right)^4 = 2 \cos^2 \alpha - 3 \cos \alpha$
 $2 \cos^2 \alpha - 3 \cos \alpha - 2 = 0$
 $\Rightarrow (2 \cos \alpha + 1)(\cos \alpha - 2) = 0 \Rightarrow \cos \alpha = \frac{-1}{2}$
 $\alpha = 2n\pi \pm \frac{2\pi}{3}$
 5. $\cos 20^\circ = K; \cos x = 2K^2 - 1$
 $= 2 \cos^2 20^\circ - 1 = \cos 40^\circ$
 on verification $x = 40^\circ$ & 320° satisfies
 6. $\Delta \geq 0 \Rightarrow \cos^2 p - 4 \sin p (\cos p - 1) \geq 0$
 $\Rightarrow \cos^2 p + 4 \sin p (1 - \cos p) \geq 0$
 $\cos^2 p \geq 0$ and $1 - \cos p \geq 0$ for all values of p
 and $\sin p \geq 0$ for $p \in (0, \pi]$

7. In $\triangle ABD, \sin A = \frac{h}{AB}$ 
 In $\triangle ABC, \cos A = \frac{AB}{2\sqrt{2}h}$
 $\frac{h}{\sin A} = 2\sqrt{2}h \cos A \Rightarrow \sin 2A = \frac{1}{\sqrt{2}}$
 $\Rightarrow A = \frac{\pi}{8}, B = \frac{3\pi}{8}$
 8. $\frac{\sin x + 1}{\cos x} = 2 \cos x$ ($\cos x \neq 0$)
 $\Rightarrow 2 \sin^2 x + \sin x - 1 = 0$
 $\Rightarrow (2 \sin x - 1)(\sin x + 1) = 0, \Rightarrow \sin x = \frac{1}{2}$
 but $\sin x + 1 \neq 0 \therefore \cos x \neq 0$
 $\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$ in $[0, 2\pi]$

9. $-\sqrt{49+25} \leq 2k+1 \leq \sqrt{49+25}$
 $-1-\sqrt{74} \leq 2k \leq -1+\sqrt{74}$
 $\frac{-1-\sqrt{74}}{2} \leq k \leq \frac{-1+\sqrt{74}}{2}$
 $k=-4,-3,-2,-1,0,1,2,3$
10. $(\cos x - \sin x)(2 \cos^2 x - \sin x \cos x - \sin^2 x) = 0$
 $\sin x = \cos x$
11. $2 \cos\left(\frac{p+q}{2}\right) \theta \cdot \cos\left(\frac{p-q}{2}\right) \theta = 0$
 $\left(\frac{p+q}{2}\right) \theta = (2n+1) \frac{\pi}{2}$
 $\theta = (2n+1) \frac{\pi}{p+q}, \quad c.d. = \frac{2\pi}{p+q}$
12. $\frac{\tan(4k+2)x - \tan(4k+1)x}{1 + \tan(4k+2)x \tan(4k+1)x} = 1$
 $\Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4}, \Rightarrow x = \frac{\pi}{4}$
 But for $x = \frac{\pi}{4}, \tan(4k+2)x$ is not defined
13. $\frac{\sin^2 \theta}{\cos \theta} + \frac{\sqrt{3} \sin \theta}{\cos \theta} = 0 \quad (\cos \theta \neq 0)$
 $\Rightarrow \sin^2 \theta + \sqrt{3} \sin \theta = 0 \Rightarrow \sin \theta (\sin \theta + \sqrt{3}) = 0$
 $\sin \theta = 0 \Rightarrow \therefore \theta = n\pi$
 $\sin \theta = -\sqrt{3}$ which is not possible
14. $\frac{3}{4} \cos 3\alpha = \frac{3\sqrt{3}}{4} \Rightarrow \cos 3\alpha = \sqrt{3} (> 1)$
 $\therefore$ There exists no solution
15. $\sin^{100} x = 1 + \cos^{100} x$
 As $\sin^{100} x$ cannot be > 1
 $\Rightarrow \cos x = 0 \Rightarrow x = (2n+1) \frac{\pi}{2} = n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$
16. $\sin x = \frac{-4 \pm \sqrt{16 - 16(a^2 - 3)}}{8}$
 $-1 \leq \frac{-1 \pm \sqrt{4 - a^2}}{2} \leq 1$
 $-2 \leq -1 \pm \sqrt{4 - a^2} \leq 2$
 $0 \leq \sqrt{4 - a^2} \leq 3$
 $4 - a^2 \geq 0 \Rightarrow a \in [-2, 2]$

17. $\sin x - i \cos 2x = \cos x - i \sin 2x$
 $\Rightarrow \sin x = \cos x$ and $\cos 2x = \sin 2x$
 No 'x' satisfies both the above equations simultaneously
18. $\tan A + \tan B = b; \tan A \tan B = c$
 $\tan(A+B) = \frac{b}{1-c} \Rightarrow \sin^2(A+B) = \left(\frac{b}{\sqrt{b^2 + (1-c)^2}} \right)^2$
19. $\sin x = \cos x$
20. Draw the graphs of, $y = 2^{\cos x}$ and $y = |\sin x|$
 No. of points of intersection is 8.
21. $\frac{|k+1|}{\sqrt{k^2+9}} \leq 1$

 LEVEL - II (H.W)

1. If $\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$
 for $0 \leq \theta \leq \pi$, then $\theta =$
 1) $\frac{\pi}{7}, \frac{5\pi}{7}, \pi$ 2) $\frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{6}, \frac{5\pi}{6}$
 3) $\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{\pi}{3}, \frac{2\pi}{3}$ 4) $\frac{\pi}{9}, \frac{4\pi}{9}, \frac{5\pi}{9}$
2. If $\alpha, \beta, \gamma, \delta$ are the four solutions of the equation $\tan\left(\theta + \frac{\pi}{4}\right) = 3 \tan 3\theta$. No two of which have equal tangents, then the value of $\tan \alpha + \tan \beta + \tan \gamma + \tan \delta =$
 1) 1 2) 0 3) -1 4) 4
3. If $0 < x, y < \frac{\pi}{2}$ then the system of equations $\sin x \cdot \sin y = 3/4$ and $\tan x \cdot \tan y = 3$ has a solution at
 1) $x = \frac{\pi}{6}, y = \frac{\pi}{6}$ 2) $x = \frac{\pi}{3}, y = \frac{\pi}{3}$
 3) $x = \frac{\pi}{12}, y = \frac{\pi}{12}$ 4) $x = \frac{\pi}{4}, y = \frac{\pi}{4}$
4. The solution set of $\sin\left(x + \frac{\pi}{4}\right) = \sin 2x$
 1) $\frac{n\pi - \left(\frac{\pi}{4}\right)}{1 + (-1)^n 2}$ 2) $\frac{n\pi + \left(\frac{\pi}{4}\right)}{1 + (-1)^n 2}$
 3) $\frac{n\pi + \left(\frac{\pi}{4}\right)}{1 - (-1)^n 2}$ 4) $\frac{n\pi - \left(\frac{\pi}{4}\right)}{1 - (-1)^n 2}$

TRIGONOMETRIC EQUATIONS

5. In a $\triangle ABC$, the angle A is greater than angle B. If the values of angles A and B satisfy the equation $3\sin x - 4\sin^3 x - k = 0$, $0 < k < 1$, then the measure of angle C =
 1) $\frac{\pi}{3}$ 2) $\frac{\pi}{2}$ 3) $\frac{2\pi}{3}$ 4) $\frac{5\pi}{6}$
6. The equation $8\cos x \cdot \cos 2x \cdot \cos 4x = \frac{\sin 6x}{\sin x}$ has a solution if
 1) $\sin x = 0$ 2) $\cos 7x = 0$ 3) $\sin 7x = 0$ 4) $\sin 8x = 0$
7. If $x \in (-\pi, \pi)$ such that
 $y = 1 + |\cos x| + |\cos^2 x| + |\cos^3 x| + \dots$
 and $8^y = 64$, then no. of values of x is
 1) 1 2) 2 3) 3 4) 4
8. If a is any real number, the number of roots of $\cot x - \tan x = a$ in the first quadrant is
 1) 2 2) 0 3) 1 4) 4
9. The number of roots of the equation $x^3 + x^2 + 2x + \sin x = 0$ in $(-2\pi, 2\pi)$
 1) 4 2) 3 3) 2 4) 1
10. If $\begin{vmatrix} \cos(A+B) & -\sin(A+B) & \cos 2B \\ \sin A & \cos A & \sin B \\ -\cos A & \sin A & \cos B \end{vmatrix} = 0$
 then B =
 1) $(2n+1)\frac{\pi}{2}$ 2) $n\pi$ 3) $(2n+1)\pi$ 4) $2n\pi$
11. If $1/6 \sin x$, $\cos x$, $\tan x$ are in G.P. then $x =$
 1) $n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$ 2) $2n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$
 3) $n\pi \pm (-1)^n \frac{\pi}{3}, n \in \mathbb{Z}$ 4) $n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$
12. The sum of the solutions of the equation $\tan x \cdot \tan 4x = 1$ for $0 < x < \pi$ is
 1) 10π 2) $\frac{3\pi}{2}$ 3) $\frac{5\pi}{2}$ 4) 2π
13. If $\tan^2 2\theta = \cot^2 \alpha$ then the general solution is
 1) $\theta = \left(\frac{1}{4}\right) \left\{ n\pi \pm \left(\frac{\pi}{2} - \alpha\right) \right\}, \forall n \in \mathbb{Z}$
 2) $\theta = \left(\frac{1}{2}\right) \left\{ n\pi \pm \left(\frac{\pi}{4} - \alpha\right) \right\}, \forall n \in \mathbb{Z}$
 3) $\theta = \left(\frac{1}{2}\right) \left\{ n\pi \pm \left(\frac{\pi}{2} - \alpha\right) \right\}, \forall n \in \mathbb{Z}$
 4) $\theta = \left(\frac{1}{4}\right) \left\{ n\pi \pm \left(\frac{\pi}{2} + \alpha\right) \right\}, \forall n \in \mathbb{Z}$

JEE-MAINS ADVANCED

14. If $\sin x + \cos x = 1 + \sin x \cdot \cos x$ then $x =$ ---
 1) $n\pi + \frac{\pi}{3}$ 2) $n\pi + (-1)^n \frac{\pi}{6}$
 3) $n\pi + (-1)^n \frac{\pi}{2} \cup 2n\pi$ 4) $n\pi$
15. The inequation $3^{\sin^2 \theta} + 3^{\cos^2 \theta} \geq 2\sqrt{3}$ is true
 1) for all real values of θ
 2) some real values of θ
 3) for imaginary values of θ
 4) for no value of θ
16. The set of values of a for which the equation $\sin^4 x + \cos^4 x = a$ has a solution is
 1) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ 2) $\left[\frac{1}{4}, 1\right]$ 3) $\left[\frac{1}{2}, 1\right]$ 4) $\left[\frac{1}{4}, \frac{1}{2}\right]$
17. If $\sin 2x \cdot \cos 2x \cdot \cos 4x = \gamma$ has a solution then γ lies in the interval
 1) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ 2) $\left[-\frac{1}{4}, \frac{1}{4}\right]$ 3) $\left[-\frac{1}{3}, \frac{1}{3}\right]$ 4) $[-1, 1]$
18. If α and β satisfying the equation $\sin \alpha + \sin \beta = \sqrt{3}(\cos \alpha - \cos \beta)$, then
 1) $\sin 3\alpha + \sin 3\beta = 1$ 2) $\sin 3\alpha + \sin 3\beta = 0$
 3) $\sin 3\alpha - \sin 3\beta = 0$ 4) $\sin 3\alpha - \sin 3\beta = 1$
19. If $\frac{3 + 2i \sin \theta}{1 - 2i \sin \theta}$ is a real number and $0 < \theta < 2\pi$, then $\theta =$
 1) π 2) $\pi/2$ 3) $\pi/3$ 4) $\pi/6$
20. The number of solutions of the equation $\sin x = x$ is
 1) 3 2) 1 3) 0 4) 2
21. The smallest positive root of the equation $\tan x - x = 0$ is in
 1) $\left(0, \frac{\pi}{2}\right)$ 2) $\left(\pi, \frac{3\pi}{2}\right)$ 3) $\left(\frac{\pi}{2}, \pi\right)$ 4) $\left(\frac{3\pi}{2}, 2\pi\right)$

LEVEL - II - KEY

- 01) 2 02) 2 03) 2 04) 4 05) 3 06) 2
 07) 4 08) 3 09) 4 10) 1 11) 2 12) 4
 13) 3 14) 3 15) 1 16) 3 17) 2 18) 2
 19) 1 20) 2 21) 2

LEVEL - II - HINTS

1. $\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0 \quad 0 \leq \theta \leq \pi$
 $\Rightarrow (\cos 6\theta + \cos 2\theta) + (1 + \cos 4\theta) = 0$
 $\Rightarrow 2 \cos 4\theta \cos 2\theta + 2 \cos^2 2\theta = 0$
 $\Rightarrow 2 \cos 2\theta (\cos 4\theta + \cos 2\theta) = 0$
 $\Rightarrow 4 \cos 2\theta \cos 3\theta \cos \theta = 0$
 $\theta = (2n+1)\frac{\pi}{2}$ or $2\theta = (2n+1)\frac{\pi}{2}$ or $3\theta = (2n+1)\frac{\pi}{2}$
 $\theta = \frac{\pi}{2}; \theta = \frac{\pi}{4}, \frac{3\pi}{4}; \theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$

2. $\frac{1 + \tan \theta}{1 - \tan \theta} = 3 \left(\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \right)$
 $\Rightarrow 3 \tan^4 \theta - 6 \tan^2 \theta + 8 \tan \theta - 1 = 0$
 $\tan \alpha + \tan \beta + \tan \gamma + \tan \delta = 0$
3. $\sin x \sin y = 3 \cos x \cos y$
 $\cos x \cdot \cos y = \frac{1}{4}$; $\sin x \cdot \sin y = \frac{3}{4}$
 $\cos(x+y) = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$
 $\cos(x-y) = \frac{4}{4} = 1$
 $x - y = 0, x + y = \frac{2\pi}{3}, x = y = \frac{\pi}{3}$
4. $\left(x + \frac{\pi}{4}\right) = n\pi + (-1)^n 2x$
 $x(1 - 2(-1)^n) = n\pi - \frac{\pi}{4}, x = \frac{n\pi - \frac{\pi}{4}}{1 - 2(-1)^n}$
5. $\sin 3x = K \quad 0 < K < 1$
 $\sin 3A = \sin(\pi - 3B) \quad (\because A > B)$
 $3A + 3B = \pi$
 $A + B = \frac{\pi}{3} \quad C = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$
6. $\sin 8x = \sin 6x, \sin x \neq 0$
 $2 \cos 7x \cdot \sin x = 0, \Rightarrow \cos 7x = 0$
7. $y = \frac{1}{1 - |\cos x|}, \frac{1}{8^{1 - |\cos x|}} = 64 = 8^2$
 $1 - |\cos x| = \frac{1}{2}, |\cos x| = \frac{1}{2} \Rightarrow \cos x = \pm \frac{1}{2}$
8. $2 \cot 2x = a \Rightarrow \cot 2x = \frac{a}{2}$
 From graph, $y = \frac{a}{2}$ intersection at only one point
 with $y = \cot 2x$
9. $x^3 + x^2 + 2x + \sin x = 0$
 only $x = 0$ satisfies the given equation
10. $\cos(A+B)[\cos(A+B)] + \sin(A+B)$
 $[\sin(A+B)] + \cos 2B(1) = 0$

- $\Rightarrow 1 + \cos 2B = 0, B = (2n+1)\frac{\pi}{2}$
11. $6 \cos^2 x = \frac{\sin^2 x}{\cos x}$
 $6 \cos^3 x + \cos^2 x - 1 = 0$
 $\cos x = \frac{1}{2}, x = 2n\pi \pm \frac{\pi}{3}$
12. $\sin x \sin 4x - \cos x \cos 4x = 0$
 $\Rightarrow \cos(x+4x) = 0 \Rightarrow \cos 5x = 0$
 $\Rightarrow 5x = (2n+1)\frac{\pi}{2}$
 $\Rightarrow x = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}$ but $x \neq \frac{5\pi}{10}$
 $\therefore$ sum of solutions 2π
13. $\tan^2 2\theta = \cot^2 \alpha = \tan^2 \left(\frac{\pi}{2} - \alpha\right)$
 $\therefore 2\theta = n\pi \pm \left(\frac{\pi}{2} - \alpha\right) \Rightarrow \theta = \frac{1}{2} \left\{ n\pi \pm \left(\frac{\pi}{2} - \alpha\right) \right\}$
14. $\sin x + \cos x - \sin x \cos x - 1 = 0$
 $(\sin x - 1)(1 - \cos x) = 0$
 $\sin x = 1 \quad \text{or} \quad \cos x = 1$
 $x = n\pi + (-1)^n \frac{\pi}{2} \quad \text{or} \quad x = 2n\pi$
15. $3^{\sin^2 \theta} = a, AM \geq GM$
 $\frac{a + \frac{3}{a}}{2} \geq \sqrt{a \times \frac{3}{a}}, a + \frac{3}{a} \geq 2\sqrt{3}$
16. $\sin^4 x + \cos^4 x = a$
 $\Rightarrow 1 - 2 \sin^2 x \cos^2 x = a \Rightarrow 1 - \frac{1}{2} \sin^2 2x = a$
 $0 \leq \sin^2 2x \leq 1 \Rightarrow 0 \leq \frac{1}{2} \sin^2 2x \leq \frac{1}{2}$
 $\frac{1}{2} \leq 1 - \frac{1}{2} \sin^2 2x \leq 1, \therefore a \in \left[\frac{1}{2}, 1\right]$
17. $\sin 4x \cos 4x = 2\gamma$
 $\sin 8x = 4\gamma \quad \Rightarrow -1 \leq 4\gamma \leq 1$
 $\Rightarrow -\frac{1}{4} \leq \gamma \leq \frac{1}{4}$

TRIGONOMETRIC EQUATIONS

18. $\frac{\sin \alpha + \sin \beta}{\cos \alpha - \cos \beta} = \sqrt{3} \Rightarrow \cot\left(\frac{\alpha - \beta}{2}\right) = -\sqrt{3}$
 $\Rightarrow \frac{\alpha - \beta}{2} = \frac{5\pi}{6} \Rightarrow 3\alpha - 3\beta = 5\pi$
 $\sin 3\alpha = \sin(5\pi + 3\beta)$
19. $\frac{(3 + 2i \sin \theta)(1 + 2i \sin \theta)}{1 + 4 \sin^2 \theta}$ is real
 Imaginary part = 0, $6 \sin \theta + 2 \sin \theta = 0$
 $\sin \theta = 0 \Rightarrow \theta = \pi$
20. $y = x, y = \sin x$
 From the graph, the only one point of intersection
21. Draw the graph of $y = \tan x$ and $y = x$



LEVEL - III

1. The number of points of intersection of the two curves $y = 2 \sin x$ and $y = 5x^2 + 2x + 3$ is
 1) 0 2) 1 3) 2 4) ∞
2. A solution (x,y) of $x^2 + 2x \sin(xy) + 1 = 0$ is
 1) (1,0) 2) $(1, 7\pi/2)$
 3) $(-2, 7\pi/2)$ 4) $(-1, 0)$
3. The solution set of equation $\cos^5 x = 1 + \sin^4 x$ is
 1) $\{n\pi, n \in I\}$ 2) $\{2n\pi, n \in I\}$
 3) $\{4n\pi, n \in I\}$ 4) $\{n\pi/2, n \in I\}$
4. Number of solution (s) of the equation $\sin x = [x]$ [where $[\bullet]$ denotes greatest integer function] is
 1) 1 2) 2 3) 0 4) infinite
5. The roots of the equation $\cos^7 x + \sin^4 x = 1$ in the interval $(-\pi, \pi)$ are
 1) $\left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$ 2) $\left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$ 3) $\left\{\frac{\pi}{2}\right\}$ 4) $\{\pi\}$
6. Let n be a positive integer such that $\sin \frac{\pi}{2n} + \cos \frac{\pi}{2n} = \frac{\sqrt{n}}{2}$ Then
 1) $n = 6$ 2) $n = 1, 2, 3, \dots, 8$
 3) $n = 5$ 4) $n = 4$

JEE-MAINS ADVANCED

7. If $x = X \cos \theta - Y \sin \theta, y = X \sin \theta + Y \cos \theta$
 a n d $x^2 + 4xy + y^2 = AX^2 + BY^2,$
 $0 \leq \theta \leq \frac{\pi}{2}, n \in Z,$ then .
 1) $\theta = \frac{\pi}{6}$ 2) $\theta = \frac{\pi}{4}$ 3) $A = 3$ 4) Both 2 and 3
8. $\sin \theta = \frac{1}{2} \left(\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} \right)$ necessarily implies
 1) $x > y$ 2) $x < y$
 3) $x = y$
 4) both x and y are purely imaginary
9. The equation $2 \cos^2 \left(\frac{x}{2} \right) \sin^2 x = x^2 + \frac{1}{x^2}, 0 \leq x \leq \frac{\pi}{2}$ has
 1) one real solution 2) no solution
 3) more than one real solution
 4) Two solutions
10. Let $2 \sin^2 x + 3 \sin x - 2 > 0$ and $x^2 - x - 2 < 0$ (x is measured in radians). Then x lies in the interval
 1) $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ 2) $\left(-1, \frac{5\pi}{6}\right)$ 3) $(-1, 2)$ 4) $\left(\frac{\pi}{6}, 2\right)$
11. If $\sin^4 x + \cos^4 y + 2 = 4 \sin x \cos y$ and $0 \leq x, y \leq \frac{\pi}{2}$ then $\sin x + \cos y$ is equal to
 1) -2 2) 0 3) 2 4) 5
12. If $0 \leq x \leq 2\pi$ and $|\cos x| \leq \sin x$, then
 1) $x \in \left[0, \frac{\pi}{4}\right]$ 2) $x \in \left[\frac{\pi}{4}, 2\pi\right]$
 3) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ 4) $[0, \pi]$
13. Number of ordered pairs (a,x) satisfying the equation $\sec^2(a+2)x + a^2 - 1 = 0;$
 $-\pi < x < \pi$ is
 1) 2 2) 1 3) 3 4) infinite
14. The number of the solutions of the equation $\cos(\pi\sqrt{x-4}) \cos(\pi\sqrt{x}) = 1$ is
 1) > 2 2) 2 3) 1 4) 0

15. The most general values of θ for which $\sin \theta - \cos \theta = \min_{a \in \mathbb{R}}(1, a^2 - 6a + 11)$ are given by

- 1) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}, n \in \mathbb{I}$
 2) $n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{4}, n \in \mathbb{I}$
 3) $2n\pi + \frac{\pi}{4}, n \in \mathbb{I}$ 4) $n\pi + \frac{\pi}{2}, n \in \mathbb{I}$

16. The number of values of x in $[0, 2\pi]$ satisfying the equation $|\cos x - \sin x| \geq \sqrt{2}$, is
 1) 0 2) 1 3) 2 4) 3

17. In $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\log_{\sin \theta}(\cos 2\theta) = 2$ has
 1) no solution 2) a unique solution
 3) two solutions 4) infinitely many solutions

18. Let $[x]$ = the greatest integer less than or equal to x and let $f(x) = \sin x + \cos x$. Then the most general solution of

$$f(x) = \left[f\left(\frac{\pi}{10}\right) \right] \text{ are}$$

- 1) $2n\pi \pm \frac{\pi}{2}, n \in \mathbb{Z}$ 2) $n\pi, n \in \mathbb{Z}$
 3) $2n\pi - \frac{\pi}{2}, n \in \mathbb{Z}$ 4) $2n\pi$ or $2n\pi + \frac{\pi}{2}$

19. Consider the solutions of the equation $\sqrt{2} \tan^2 x - \sqrt{10} \tan x + \sqrt{2} = 0$ in the range $0 \leq x \leq \pi/2$. Then only one of the following is true

- 1) no solutions exist for x in the given range
 2) two solutions x_1 and x_2 exist with $x_1 + x_2 = \pi/4$
 3) two solutions x_1 and x_2 exist with $x_1 + x_2 = \pi/6$
 4) two solutions x_1 and x_2 exist with $x_1 + x_2 = \pi/2$

20. The number of solutions of the equation $|\cos x| = 2[x]$ is (where $[\bullet]$ denotes the greatest integer function)
 1) 0 2) 1 3) 2 4) 3

21. If α, β are solutions of $\sin^2 x + a \sin x + b = 0$ and $\cos^2 x + c \cos x + d = 0$ then $\sin(\alpha + \beta)$ equals

1) $\frac{2ac}{a^2 + c^2}$ 2) $\frac{a^2 + c^2}{2ac}$ 3) $\frac{2bd}{b^2 + d^2}$ 4) $\frac{b^2 + d^2}{2bd}$

22. If $\sqrt{3} \sin \pi x + \cos \pi x = x^2 - \frac{2}{3}x + \frac{19}{9}$, then x is equal to

1) $-\frac{1}{3}$ 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $\frac{4}{3}$

23. The equation $1 + \sin^2 ax = \cos x$ has a unique solution then a is

- 1) rational 2) irrational
 3) integer 4) whole number

24. If n be the number of solutions of the equation

$$|\cot x| = \cot x + \frac{1}{\sin x} \quad (0 < x < 2\pi), \text{ then } n =$$

- 1) 1 2) 2 3) 3 4) 4

25. If $\frac{1 - \tan x}{1 + \tan x} = \tan y$ and $x - y = \frac{\pi}{6}$, then x, y are respectively

- 1) $\frac{5\pi}{24}, \frac{\pi}{24}$ 2) $-\frac{7\pi}{24}, -\frac{11\pi}{24}$
 3) $-\frac{115\pi}{24}, -\frac{119\pi}{24}$ 4) All the above

26. If $\theta \in [0, 5\pi]$ and $r \in \mathbb{R}$ such that $2 \sin \theta = r^4 - 2r^2 + 3$ then the maximum number of values of the pair (r, θ) is

- 1) 8 2) 10 3) 6 4) 4

27. If $\frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta - \cos \theta} - \frac{\cos \theta}{\sqrt{1 + \cot^2 \theta}}$

$$-2 \tan \theta \cot \theta = -1, \theta \in [0, 2\pi] \text{ then}$$

- 1) $\theta \in \left(0, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}\right\}$ 2) $\theta \in \left(\frac{\pi}{2}, \pi\right) - \left\{\frac{3\pi}{4}\right\}$
 3) $\theta \in \left(\pi, \frac{3\pi}{2}\right) - \left\{\frac{5\pi}{4}\right\}$ 4) $\theta \in (0, \pi) - \left\{\frac{\pi}{4}, \frac{\pi}{2}\right\}$

TRIGONOMETRIC EQUATIONS

28. The least difference between the roots, in the first quadrant $\left(0 \leq x \leq \frac{\pi}{2}\right)$, of the equation $4 \cos x (2 - 3 \sin^2 x) + (\cos 2x + 1) = 0$, is
- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$
29. The set of all x in $(-\pi, \pi)$ satisfying $|4 \sin x - 1| < \sqrt{5}$ is given by
- 1) $\left(-\frac{3\pi}{10}, \frac{3\pi}{10}\right)$ 2) $\left(-\frac{\pi}{10}, \pi\right)$
- 3) $(-\pi, \pi)$ 4) $\left(-\pi, \frac{3\pi}{10}\right)$
30. The general solution of the equation $\frac{1 - \sin x + \dots + (-1)^n \sin^n x + \dots}{1 + \sin x + \dots + \sin^n x + \dots} = \frac{1 - \cos 2x}{1 + \cos 2x}$ is
- 1) $(-1)^n \left(\frac{\pi}{3}\right) + n\pi, \forall n \in I$
- 2) $(-1)^n \left(\frac{\pi}{6}\right) + n\pi, \forall n \in I$
- 3) $(-1)^{n+1} \left(\frac{\pi}{6}\right) + n\pi, \forall n \in I$
- 4) $(-1)^{n-1} \left(\frac{\pi}{3}\right) + n\pi, \forall n \in I$
31. The value of x between 0 and 2π which satisfy the equation $\sin x \sqrt{8 \cos^2 x} = 1$ are in AP with common difference
- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{8}$ 3) $\frac{3\pi}{8}$ 4) $\frac{5\pi}{8}$
32. The no. of values of x in the interval $[0, 3\pi]$ satisfying the equation $2 \sin^2 x + 5 \sin x - 3 = 0$ is
- 1) 1 2) 4 3) 6 4) 2
33. The solution set satisfying $\tan x > 1$ is
- 1) $\left(n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{2}\right)$ 2) $\left(n\pi + \frac{\pi}{4}, n\pi\right)$
- 3) $\left(n\pi + \frac{\pi}{4}, \infty\right)$ 4) ϕ

JEE-MAINS ADVANCED

34. $0 \leq a \leq 3, 0 \leq b \leq 3$ and the equation, $x^2 + 4 + 3 \cos(ax + b) = 2x$ has atleast one solution then the value of $a + b$
- 1) $\frac{\pi}{2}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) π
35. If $2 \tan^2 x - 5 \sec x = 1$ for exactly 7 distinct values of $x \in \left[0, \frac{n\pi}{2}\right], n \in N$ then the greatest value of n is
- 1) 13 2) 17 3) 19 4) 15
36. If the equation $\cot^4 x - 2 \sec^2 x + a^2 = 0$ has at least one solution, then the sum of all possible integral values of a is equal to
- 1) 4 2) 3 3) 2 4) 0
37. The equation $\sin^4 x + \cos^4 x + \sin 2x + \alpha = 0$ is solvable for
- 1) $-\frac{5}{2} \leq \alpha \leq \frac{1}{2}$ 2) $-3 \leq \alpha \leq 1$
- 3) $-\frac{3}{2} \leq \alpha \leq \frac{1}{2}$ 4) $-1 \leq \alpha \leq 1$
38. The number of solution of $\sum_{r=1}^5 \cos rx = 5$ in the interval $[0, 2\pi]$ is
- 1) 0 2) 1 3) 5 4) 2
39. The number of solution of $x \in [0, 2\pi]$ for which $[\sin x + \cos x] = 3 + [-\sin x] + [-\cos x]$ (where $[.]$ denotes the greatest integer function) is
- 1) 0 2) 4 3) infinite 4) 1
40. The equation $\cos^8 x + b \cos^4 x + 1 = 0$ will have a solution if b belongs to
- 1) $(-\infty, 2]$ 2) $[2, \infty)$ 3) $(-\infty, -2]$ 4) $[1, \infty)$

LEVEL - III-KEY

- 01) 1 02) 2 03) 2 04) 2 05) 1 06) 1
 07) 4 08) 3 09) 2 10) 4 11) 3 12) 3
 13) 3 14) 3 15) 2 16) 3 17) 2 18) 4
 19) 4 20) 1 21) 1 22) 2 23) 2 24) 1
 25) 4 26) 3 27) 4 28) 1 29) 1 30) 2
 31) 1 32) 2 33) 1 34) 4 35) 4 36) 4
 37) 3 38) 4 39) 3 40) 3

LEVEL - III-HINTS

- We have,

$$y = 5x^2 + 2x + 3 = 5\left[x^2 + \frac{2}{5}x + \frac{3}{5}\right]$$

$$= 5\left[\left(x + \frac{1}{5}\right)^2 + \frac{3}{5} - \frac{1}{25}\right] = 5\left(x + \frac{1}{5}\right)^2 + \frac{14}{5} > 2$$

since $y = 2\sin x \leq 2$, so there cannot be any point of intersection.
- Verify from the options
- $\cos^5 x = 1 + \sin^4 x$ Minimum value of R.H.S. is 1 and maximum value of L.H.S. is 1. So equality holds only when both sides are simultaneously 1. i.e., $\cos x = 1$ and $\sin x = 0$
 $\cos x = 1 \Rightarrow x = 2n\pi$ and $\sin x = 0 \Rightarrow x = n\pi$
 So, solution set $x = 2n\pi$.
- $y = \sin x = [x]$ from graph there are two solutions. $x = 0$ and $x = \frac{\pi}{2}$
- $\cos^7 x + \sin^4 x = 1$ in the interval $(-\pi, \pi)$. In L.H.S. one term has to be one and one term is 0, otherwise it is not possible.
 $\cos^7 x = 0, \sin^4 x = 1 \Rightarrow x = \pm \frac{\pi}{2}$
 or $\cos^7 x = 1, \sin x = 0 \Rightarrow x = 0$
- Verification by $n=6$.
- Let $E = x^2 + 4xy + y^2$
 Substituting respective values, we have
 $E = (1 + 4\sin\theta\cos\theta)X^2$
 $+ 4(\cos^2\theta - \sin^2\theta)XY$
 $+ (1 - 4\sin\theta\cos\theta)Y^2$
 $\Rightarrow (1 + 2\sin 2\theta)X^2 + 4\cos(2\theta)XY + (1 - 2\sin 2\theta)Y^2$
 $= AX^2 + BY^2$
 $\Rightarrow \theta = \frac{\pi}{4}$, and $A = 1 + 2 = 3, B = 1 - 2 = -1$
- $\sin\theta \leq 1$
- Since $x^2 + x^{-2} = (x - x^{-1})^2 + 2 \geq 2$ and
 $2\cos^2 \frac{x}{2} \sin^2 x \leq 2$.
 $\therefore$ the given equation is valid only if

- $$2\cos^2 \frac{x}{2} \sin^2 x = 2$$
- $$\Rightarrow \cos \frac{x}{2} = \sin x = 1, \text{ which cannot be true}$$
- we have $2\sin^2 x + 3\sin x - 2 > 0$
 $\Rightarrow (2\sin x - 1)(\sin x + 2) > 0$
 $\Rightarrow 2\sin x - 1 > 0 [\because \sin x + 2 > 0 \forall x \in R]$
 $\Rightarrow \sin x > 1/2 \Rightarrow x \in (\pi/6, 5\pi/6)$ Also \
 $x^2 - x - 2 < 0$
 $\Rightarrow (x - 2)(x + 1) < 0 \Rightarrow -1 < x < 2$.
 As $2 < \frac{5\pi}{6}$, we obtain that x must lie in $\left(\frac{\pi}{6}, 2\right)$
 - We have $\sin^4 x + \cos^4 y + 2 = 4\sin x \cos y$
 $\Rightarrow \sin^4 x + \cos^4 y + 2 - 4\sin x \cos y = 0$
 $\Rightarrow (\sin^2 x - 1)^2 + (\cos^2 y - 1)^2 + 2\sin^2 x$
 $+ 2\cos^2 y - 4\sin x \cos y = 0$
 $\Rightarrow (\sin^2 x - 1)^2 + (\cos^2 y - 1)^2 + 2(\sin x - \cos y)^2 = 0$
 Which is true if $\sin^2 x = 1, \cos^2 y = 1$
 $\sin x = \cos y$. As, $0 \leq x, y \leq \pi/2$. we get
 $\sin x = \cos y = 1 \Rightarrow \sin x + \cos y = 2$
 - We have,
 $|\cos x| \leq \sin x \Rightarrow \sin x \geq 0 (\because |\cos x| \geq 0)$
 $\Rightarrow x \notin (\pi, 2\pi)$ If $x = 2\pi, |\cos 2\pi| \leq \sin 2\pi$
 which is not possible $\therefore x \in \left[0, \frac{\pi}{2}\right]$, then
 $|\cos x| \leq \sin x \Rightarrow x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$
 If $x \in \left(\frac{\pi}{2}, \pi\right)$, then $|\cos x| \leq \sin x$
 $\Rightarrow -\cos x \leq \sin x (\because \cos x < 0)$
 $\Rightarrow \tan x \leq -1 \Rightarrow x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right]$
 $\Rightarrow x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right] \cup \left(\frac{\pi}{2}, \frac{3\pi}{4}\right]$

TRIGONOMETRIC EQUATIONS

13. Given equation is $\sec^2(a+2)x + a^2 - 1 = 0$
 $\Rightarrow \tan^2(a+2)x + a^2 = 0$
 $\Rightarrow \tan^2(a+2)x = 0$ and $a = 0$
 $\Rightarrow \tan^2 2x = 0 \Rightarrow x = 0, \frac{\pi}{2}, -\frac{\pi}{2}$
 $\therefore (0, 0), (0, \pi/2), (0, -\pi/2)$ are ordered pairs satisfying the equation.
14. Clearly, $x \geq 4$ (Since $\sqrt{x-4}$ is real)
 so that $\sqrt{x}$ is also real. Again,
 if $\cos(\pi\sqrt{x}) < 1$ then
 $\cos(\pi\sqrt{x-4}) > 1$ (since their product = 1) But both of these are not possible (since $\cos \theta$ cannot be greater than 1) $\therefore \cos(\pi\sqrt{x-4}) = 1$ and $\cos(\pi\sqrt{x}) = 1$
 $\therefore x-4 = 0$ and $x = 0$
 But $x = 0$ is not possible $\therefore x = 4$ is the only solution
15. We have $\sin \theta - \cos \theta = \min_{a \in \mathbb{R}} \{1, a^2 - 6a + 11\}$
 Since $a^2 - 6a + 11 = (a-3)^2 + 2 \geq 2$ for all a
 $\therefore \sin \theta - \cos \theta = 1$
 $\Rightarrow \sin\left(\theta - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4}$
 $\Rightarrow \theta - \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}$
 $\therefore \theta = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{4}$ where $n \in \mathbb{Z}$
16. Given equation is $|\cos x - \sin x| \geq \sqrt{2}$
 Since $|\cos x - \sin x| \leq \sqrt{1+1} = \sqrt{2}$
 $\therefore$ we must have $|\cos x - \sin x| = \sqrt{2}$
 $\Rightarrow \left|\cos\left(x + \frac{\pi}{4}\right)\right| = 1 \Rightarrow \cos\left(x + \frac{\pi}{4}\right) = 1, -1$
 $\therefore x = \frac{3\pi}{4}, \frac{7\pi}{4}$

JEE-MAINS ADVANCED

17. We have, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$
 $\therefore -1 \leq \sin \theta \leq 1$, here $0 < \sin \theta < 1$. Now,
 $\log_{\sin \theta} \cos 2\theta = 2 \Rightarrow 1 - 2\sin^2 \theta = \sin^2 \theta$
 $\Rightarrow \sin \theta = \frac{1}{\sqrt{3}}$
 The given equation has a unique solution
18. Here $f\left(\frac{\pi}{10}\right) = \sin 18^\circ + \cos 18^\circ$
 $= \sqrt{2} \sin(45^\circ + 18^\circ) = \sqrt{2} \sin 63^\circ$
 Since $\sin 63^\circ > \sin 45^\circ = \frac{1}{\sqrt{2}}$ and $\sqrt{2} \sin 63^\circ > 1$,
 $\therefore 1 < f\left(\frac{\pi}{10}\right) < \sqrt{2} \Rightarrow \left[f\left(\frac{\pi}{10}\right)\right] = 1$
 $\therefore$ The equation is $\sin x + \cos x = 1$
 $\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \cos \frac{\pi}{4}$
 $\Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$
19. $\tan x_1 + \tan x_2 = \sqrt{5}$, $\tan x_1 \tan x_2 = 1$
20. $|\cos x| = 2[x] = y$, $y = |\cos x|$ and $y = 2[x]$
 Graph of $|\cos x|$ and $2[x]$ don't cut each other for any real value of x . Hence number of solutions is zero.
21. $\sin \alpha + \sin \beta = -a$, $\cos \alpha + \cos \beta = -c$
 Apply transformations and on dividing
 $\tan \frac{\alpha + \beta}{2} = \frac{a}{c}$, $\sin(\alpha + \beta) = \frac{2ac}{a^2 + c^2}$
22. Verify the options
23. $1 + \sin^2 ax = \cos x$
 Minimum value of L.H.S is 1 and maximum value of R.H.S is 1, equality exist only when both sides
 $\Rightarrow \sin^2 ax = 0$ and $\cos x = 1$
 are 1 $\Rightarrow ax = n\pi$ and $x = 2n\pi$
 If a is rational, there are infinite or no common solution but when a is irrational only one common solution i.e., $x = 0$

$$24. \quad |\cot x| = \begin{cases} -\cot x, & \cot x < 0 \\ \cot x, & \cot x \geq 0 \end{cases}$$

CASE I: $\cot x \geq 0$

$$\cot x = \cot x + \frac{1}{\sin x} \Rightarrow \frac{1}{\sin x} = 0$$

$\Rightarrow \sin x = \infty$, This is not possible.

CASE II: $\cot x < 0$

$$\Rightarrow -\cot x = \cot x + \frac{1}{\sin x}$$

$$\therefore \cos x = -\frac{1}{2} = -\cos \frac{\pi}{3}$$

$$\Rightarrow \cos x = \cos \left(\pi + \frac{\pi}{3} \right) \text{ or } \cos \left(\pi - \frac{\pi}{3} \right)$$

$$\therefore x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\text{But, } x = \frac{4\pi}{3} \Rightarrow \cot x > 0, \therefore n = 1$$

$$25. \quad \tan \left(\frac{\pi}{4} - x \right) = \tan y = \tan \left(x - \frac{\pi}{6} \right)$$

$$\left\{ \because x - y = \frac{\pi}{6} \right\}$$

$$\therefore \frac{\pi}{4} - x = n\pi + x - \frac{\pi}{6} \Rightarrow \left(\frac{5\pi}{12} - n\pi \right) = 2x$$

$$\Rightarrow x = (5 - 12n) \frac{\pi}{24} \text{ and } y = x - \frac{\pi}{6} \quad \text{Options}$$

(1)(2)(3) correspond to $n = 0, 1, 10$ respectively.

26. But max value of $2\sin \theta$ is 2 and it is attained at r

$$= 1, -1 \Rightarrow \theta = n\pi + (-1)^n \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2} \left\{ \because \theta \in [0, 5\pi] \right\}$$

$\therefore$ The number of values of the pair

$$(r, \theta) = 2 \times 3 = 6$$

$$27. \quad \sin \theta - \cos \theta \neq 0, \therefore \theta \neq \frac{\pi}{4}, \frac{5\pi}{4}$$

$$\text{Now } 1 + \cos \theta (\sin \theta - |\sin \theta|) - 2 = -1$$

$$\cos \theta (\sin \theta - |\sin \theta|) = 0$$

$$\therefore \theta \in (0, \pi) - \left\{ \frac{\pi}{4}, \frac{\pi}{2} \right\}$$

$$28. \quad \text{we have } 4\cos x (3\cos^2 x - 1) + 2\cos^2 x = 0$$

$$2\cos x (3\cos x + 2)(2\cos x - 1) = 0$$

either $\cos x = 0$ which gives

$$x = \pi/2 \text{ or } \cos x = -2/3$$

Which gives no values of x for which

$$0 \leq x \leq \pi/2 \text{ or } \cos x = 1/2$$

which gives $x = \pi/3$

$$\text{So, the required difference} = \frac{\pi}{6}$$

$$29. \quad \text{We have, } |4\sin x - 1| < \sqrt{5}$$

$$\Rightarrow -\left(\frac{\sqrt{5}+1}{4} \right) < \sin x < \frac{\sqrt{5}+1}{4}$$

$$\Rightarrow \sin \left(\frac{-3\pi}{10} \right) < \sin x < \sin \left(\frac{3\pi}{10} \right)$$

30. The equation

$$\frac{1 - \sin x + \dots + (-1)^n \sin^n x + \dots}{1 + \sin x + \dots + \sin^n x + \dots} = \frac{1 - \cos 2x}{1 + \cos 2x}$$

$$\Rightarrow \frac{1}{1 + \sin x} \cdot \frac{1 - \sin x}{1} = \frac{2\sin^2 x}{2\cos^2 x}$$

$$\Rightarrow 2\sin^2 x + \sin x - 1 = 0$$

$$\Rightarrow \sin x = \frac{-1 \pm \sqrt{1+8}}{4} = \frac{-1 \pm 3}{4}$$

$$\Rightarrow \sin x = -1 \text{ or } \sin x = \frac{1}{2}$$

Since, $\sin x \neq -1$, we have

$$\sin x = \frac{1}{2} = \sin \left(\frac{\pi}{6} \right),$$

$$\therefore x = n\pi + (-1)^n \frac{\pi}{6}$$

TRIGONOMETRIC EQUATIONS

31. $\sin 2x = \pm \frac{1}{\sqrt{2}}$

32. Given equation is

$$2\sin^2 x + 5\sin x - 3 = 0$$

$$\Rightarrow (\sin x + 3)(2\sin x - 1) = 0$$

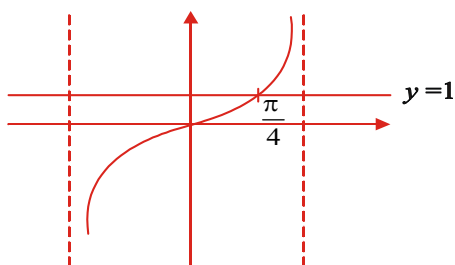
$$\Rightarrow \sin x \neq -3 \text{ or } \sin x = \frac{1}{2}$$

given $0^\circ \leq x \leq 540^\circ$, $\therefore$ no. of values are 4.

i.e., $30^\circ, 150^\circ, 390^\circ, 510^\circ$

33. Graph of $y = \tan x$ is as shown

$\therefore$ For $\tan x > 1$



$$n\pi + \frac{\pi}{4} < x < n\pi + \frac{\pi}{2}$$

34. $x^2 - 2x + 4 = -3\cos(ax + b)$

$$\Rightarrow (x-1)^2 + 3 = -3\cos(ax + b)$$

As $-1 \leq \cos(ax + b) \leq 1$ and $(x-1)^2 \geq 0$

$\therefore$ equation (i) is only possible if,

$$\cos(ax + b) = -1 \text{ and } (x-1) = 0 \text{ so}$$

$$a + b = \pi, (\text{where } a + b \leq 6)$$

35. $\sec x = 3 \Rightarrow \cos x = \frac{1}{3}$

Which gives two values of x in each of

$[0, 2\pi], (2\pi, 4\pi], (4\pi, 6\pi]$ and one value in

$$6\pi + \frac{3\pi}{2} = 15\frac{\pi}{2} \therefore \text{Greatest value of } n = 15$$

36. $\cot^4 x - 2\cot^2 x - 2 + \alpha = 0$

$$(\cot^2 x - 1)^2 = 3 - \alpha, 3 - \alpha \geq 0, \alpha = -1, 0, 1$$

JEE-MAINS ADVANCED

37. $\sin^4 x + \cos^4 x + \sin 2x + \alpha = 0$

Let $\sin 2x = y$. then the given equation becomes

$$y^2 - 2y - 2(1 + \alpha) = 0 \text{ where } -1 \leq y \leq 1$$

($\because -1 \leq \sin 2x \leq 1$), for real, discriminant ≥ 0

38. $\sum_{r=1}^5 \cos rx = 5 \Rightarrow \cos x + \cos 2x + \cos 3x$

$$+ \cos 4x + \cos 5x = 5$$

Which is possible only when

$$\cos x = \cos 2x = \cos 3x$$

$$= \cos 4x = \cos 5x = 1 \text{ and is satisfied by } x=0 \text{ and}$$

$$x = 2\pi.$$

39. $[\sin x + \cos x] = 3 + [-\sin x] + [-\cos x]$

Maximum value of left - hand side is 1 and minimum of right hand side is also 1

$$\Rightarrow [\sin x + \cos x] = 3 + [\sin x] + [-\cos x] = 1$$

$$\Rightarrow [\sin x + \cos x] = 1,$$

$$[-\sin x] = -1, [-\cos x] = -1$$

40. $\cos^8 x + b \cos^4 x + 1 = 0$

$$\Rightarrow b = -\left(\cos^4 x + \frac{1}{\cos^4 x}\right) \leq -2, \forall x \in R$$

$$\Rightarrow b \in (-\infty, -2]$$



LEVEL - IV

1. **Assertion (A):** $3 \sin x + 4 \cos x = 7$ has no solution

Reason (R): $a \cos x + b \sin x = c$ has no

solution if $|c| > \sqrt{a^2 + b^2}$

1) Both A and R are true and R is correct explanation of A

2) Both A and R are true and R is not the correct explanation of A

3) A is true but R is false

4) A is false but R is true

2. **Statement I:** Principal value of $\cos \theta = -1$ is π

Statement II: Principal value of $\sin \theta = 0$ is π

Which of the above statement is correct?

- 1) Only I 2) Only II
3) Both I and II 4) Neither I or II

3. **Statement I:** The set of values of x for which

$$\frac{\tan 3x - \tan 2x}{1 + \tan 3x \tan 2x} = 1 \text{ is } \left\{ n\pi + \frac{\pi}{4}, n \in I \right\}$$

Statement II: The expression

$$(1 + \tan x + \tan^2 x)(1 - \cot x + \cot^2 x)$$

is positive for all defined values of x

III) $e^{\sin x} - e^{-\sin x} \neq 4$ for any real values of x .

Which of the following is correct?

- 1) only I, II 2) only II, III
3) only I, III 4) I, II, III

4. **Statement I:** If $\cot \left(\frac{\pi}{3} \cos 2\pi x \right) = \sqrt{3}$, then

the general solution of the equation is

$$x = n \pm \frac{1}{6}, n \in I.$$

Statement II: If $\tan P \theta = \tan Q \theta$, then the values of θ from an A.P with common

difference $\frac{\pi}{P-Q}$.

Which of the above statements are correct?

- 1) only I 2) only II
3) Both I and II 4) neither I or II

5. **Statement I:** If x lies in the 1st Quadrant and

$$\cos x + \cos 3x = \cos 2x \text{ then } x = 30^\circ \text{ or } 45^\circ$$

Statement II: $x \in (0, 2\pi)$ and $\operatorname{cosec} x + 2 = 0$

$$\text{then } x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

Statement III: $x \in [0, 2\pi]$ and

$$(2 \cos x - 1)(3 + 2 \cos x) = 0 \text{ then } x = \frac{\pi}{3}, \frac{5\pi}{3}$$

Which of the above statements are correct?

- 1) only I, II 2) only II, III
3) only I, III 4) I, II, III

LEVEL - IV-KEY

- 01) 1 02) 1 03) 2 04) 3 05) 2

LEVEL - IV-HINTS

1. Assertion is true since $|c| > \sqrt{a^2 + b^2}$

2. By range of principal solutions

3. I. for $x = \frac{\pi}{4}$, $\tan 2x$ is undefined

$$\text{II. G.E.} = 1 + \cot^2 x + \tan^2 x > 0$$

$$\text{III. If } e^{\sin x} - e^{-\sin x} = 4 \text{ then}$$

$$\sin x = \log_e (2 + \sqrt{5}) > 1 \text{ which is not true}$$

4. I. $\cos 2\pi x = \cos \frac{\pi}{3} \Rightarrow 2\pi x = 2n\pi \pm \frac{\pi}{3}$

$$\Rightarrow x = n \pm \frac{1}{6}, n \in \mathbb{Z}$$

$$\text{II. } P\theta = n\pi + Q\theta \Rightarrow \theta = \frac{n\pi}{P-Q}$$

$$\therefore \text{common difference} = \frac{\pi}{P-Q}$$

5. I. $\cos x + \cos 3x = \cos 2x$

$$\Rightarrow 2 \cos 2x \cos x = \cos 2x$$

$$\Rightarrow \cos 2x = 0 \text{ or } 2 \cos x = 1$$

$$\therefore x = 45^\circ \text{ or } x = 60^\circ$$

$$\text{II. } \operatorname{cosec} x = -2$$

$$\therefore x = \frac{7\pi}{6}, \frac{11\pi}{6} \text{ satisfies}$$

$$\text{III. } (2 \cos x - 1)(3 + 2 \cos x) = 0$$

$$\Rightarrow \cos x = \frac{1}{2} \text{ or } \cos x = \frac{-3}{2}$$

(which is not possible)

$$\therefore x = \frac{\pi}{3}, \frac{5\pi}{3}$$